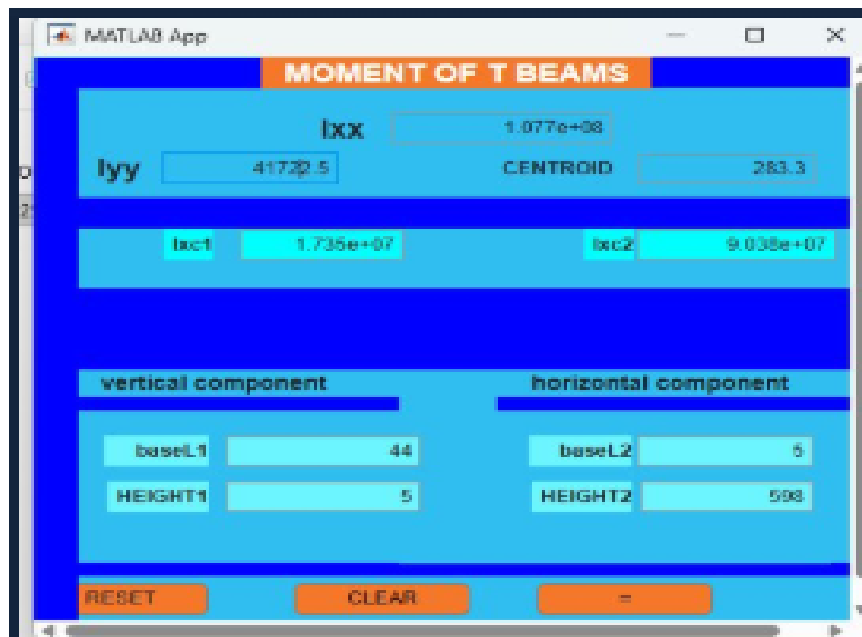


MATLAB T-Beam Moment of Inertia Application Report

This report documents the development of a MATLAB application designed to calculate the First Moment of Area and Second Moment of Area (Moment of Inertia) for T-beam sections. The project provided practical experience in MATLAB programming, engineering calculations, graphical user interface (GUI) development, and numerical problem solving.

Application Screenshot



Project Objective

The objective of this project was to create a user-friendly MATLAB application capable of calculating geometric properties of T-beams. Users can input beam dimensions and obtain centroid location, first moment of area, and second moment of area values quickly and accurately.

Background Theory

The First Moment of Area is used in engineering applications such as shear stress calculations. The Second Moment of Area (Moment of Inertia) measures a section's resistance to bending. For T-beams, these properties are determined by dividing the shape into simpler rectangles, locating the centroid, and applying the parallel axis theorem.

MATLAB Skills Learned

During the development of this application, the following MATLAB skills were learned:

- Designing graphical user interfaces using MATLAB App Designer.
- Creating input fields, labels, buttons, and output displays.
- Writing callback functions for user interaction.
- Implementing engineering equations in MATLAB code.

- Performing numerical calculations efficiently.
- Debugging and testing programs.
- Improving user experience through interface design.

Application Features

- Input of T-beam dimensions.
- Automatic centroid calculation.
- Calculation of first moment of area.
- Calculation of second moment of area about both principal axes.
- Clear and reset functions for easy reuse.
- Instant display of results.

Challenges Encountered

Several challenges were encountered, including implementing the correct engineering formulas, handling user input validation, ensuring accurate centroid calculations, and presenting results clearly within the GUI. These challenges improved problem-solving and debugging skills.

Results and Evaluation

The completed application successfully calculates T-beam section properties and provides results through an intuitive graphical interface. The program reduces manual calculation effort and minimizes computational errors.

Conclusion

This project demonstrated how MATLAB can be used to solve real engineering problems through software development. The application combines engineering theory with programming techniques, resulting in a practical tool for structural analysis. The project strengthened understanding of moments of inertia, centroid calculations, GUI development, and MATLAB programming practices.